TEC | Tecnológico de Costa Rica

Computer Engineering Department
Computer Engineering
IC6400 - Operations Research
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Simplex Method Project

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George Dantzig

- George Bernard Dantzig (1914-2005) was an American mathematician and computer scientist.
- He is best known for his development of the Simplex algorithm, a method for solving linear programming problems.
- Dantzig's work in optimization and operations research has had a profound impact on various fields, including economics, engineering, and logistics.
- He made significant contributions to the theory of linear programming, integer programming, and network flow problems.
- Dantzig's work has been widely recognized and has earned him numerous awards and honors throughout his career.

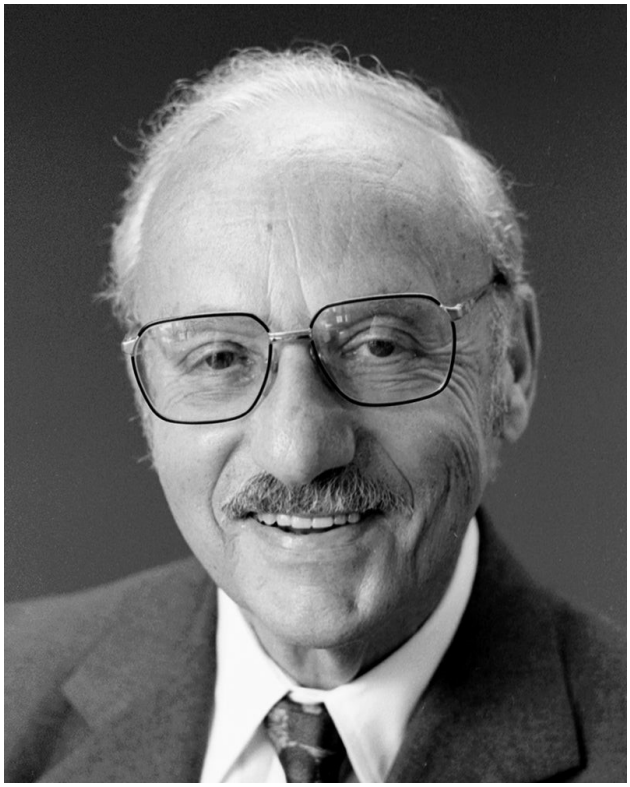


Figure 1: Image inspired by George Dantzig, the creator of the Simplex Algorithm.

The Simplex Algorithm

- It is a "mechanical" method that allows to find optimal solutions to linear programming problems.
- It was invented by George Dantzig in 1947.
- Uses matrix operations until it finds the optimal solution or determines that the problem is unbounded or infeasible.
- It is based in a geometric interpretation of the problem, where the feasible region is a convex polytope and the optimal solution is found at one of its vertices.

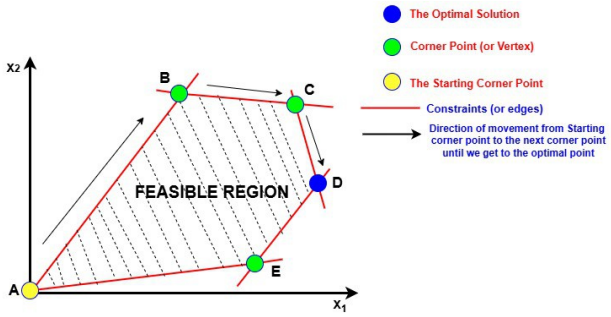


Figure 2: Image inspired by the geometric interpretation of the Simplex Algorithm.

Original Problem

Description

- Problem Name: Problema 9
- Amount of decision variables: 2
- Amount of restrictions: 3
- Total Amount of variables (decision + slack): 5
- Type of problem: Minimization

Equations

$$
\begin{aligned}
&\text{Min } Z = 4\,x_1 - x_2 \\
&\text{subject to:} \\
&\qquad 2\,x_1 + x_2 \leq 8 \\
&\qquad x_2 \leq 5 \\
&\qquad x_1 - x_2 \leq 4 \\
&\qquad x_1, x_2 \geq 0
\end{aligned}
$$

Initial Table

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	-4.0000	1.0000	0.0000	0.0000	0.0000	0.0000
0.0000	2.0000	1.0000	1.0000	0.0000	0.0000	8.0000
0.0000	0.0000	1.0000	0.0000	1.0000	0.0000	5.0000
0.0000	1.0000	-1.0000	0.0000	0.0000	1.0000	4.0000

Intermediate Tables

Iteration 1

Entering variable: x_2 (most positive coefficient in the Z row: 1.0000). Leaving variable: s_2 (row 2, minimum ratio 5.0000).

Finding the Pivot

Z	x_1	x_2	s_1	s_2	s_3	b	P
1.0000	-4.0000	1.0000	0.0000	0.0000	0.0000	0.0000	-
0.0000	2.0000	1.0000	1.0000	0.0000	0.0000	8.0000	8.0000
0.0000	0.0000	**1.0000**	0.0000	1.0000	0.0000	5.0000	5.0000
0.0000	1.0000	-1.0000	0.0000	0.0000	1.0000	4.0000	-

Canonize the Pivot column with Pivot value

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	-4.0000	0.0000	0.0000	-1.0000	0.0000	-5.0000
0.0000	2.0000	0.0000	1.0000	-1.0000	0.0000	3.0000
0.0000	0.0000	1.0000	0.0000	1.0000	0.0000	5.0000
0.0000	1.0000	0.0000	0.0000	1.0000	1.0000	9.0000

Final Table

Z	x_1	x_2	s_1	s_2	s_3	b
1.0000	-4.0000	0.0000	0.0000	-1.0000	0.0000	-5.0000
0.0000	2.0000	0.0000	1.0000	-1.0000	0.0000	3.0000
0.0000	0.0000	1.0000	0.0000	1.0000	0.0000	5.0000
0.0000	1.0000	0.0000	0.0000	1.0000	1.0000	9.0000

Solution

Solution

The optimal solution is:

- $x_1 = 0.0000$ (non-basic)
- $x_2 = 5.0000$
- $s_1 = 3.0000$
- $s_2 = 0.0000$ (non-basic)
- $s_3 = 9.0000$

The optimal value of the objective function is: $Z = -5.0000$